Vakint reduction benchmark

RustRed versus alphaLoop, MATAD and FMFT · reduction graphs from one to five loops · precision of the results

2026-11-07 · AMD EPYC 9754 128-Core Processor · one core per timed process except 10 marked cells · vakint 0.2.0

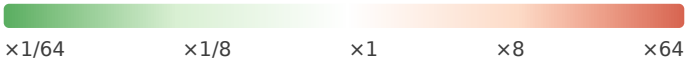
Reading guide. Bold: RustRed median, then the reference's (M = MATAD, F = FMFT, αL = alphaLoop) and the ratio RustRed ÷ reference; green = RustRed faster, red = slower (log scale); (·) = minimum over the runs. *Reduction*: masters retained; *full*: complete call with master values; *warm*: the identical call repeated in the process (step cache filled). Every timed call follows an untimed warm-up on another integral, so one-time setup is excluded (listed below the table). *IBP*: RustRed's time in the reduction itself. Five loops (no FORM backend): a master is recognised in any form and returned without a reduction walk; other inputs run RustRed's modular ordered walk (`modular_sampled`, on the cores a cell names).

Timings

Table 1: **RustRed versus the FORM backends** (MATAD and alphaLoop: one to three loops; FMFT: four loops; none reaches five). Terminals: distinct RustRed master symbols in the retained result (one to three loops: terminals of the closed program, symmetric copies included). Agreement: digits shared with RustRed at $m = \mu = 1$, worst order of ε.

Test case	Terminals	Reduction vs M / F	Reduction vs αL	Warm reduction	Full call vs M / F	Full call vs αL	Agreement
One loop							
QCD ladder, q = 0 quark loop, no rung	1	**4.35 ms** (3.24) M 24.1 ms (23.2) **×0.18** IBP 1.51 ms	**4.35 ms** (3.24) αL 22.5 ms (22.0) **×0.19**	**4.44 ms** (3.19) cold 4.35 ms **×1.0**	**23.3 ms** (18.6) M 160 ms (158) **×0.15**	**23.3 ms** (18.6) αL 39.6 ms (39.3) **×0.59**	**31** digits
QCD ladder, O(q²) quark loop, no rung	1	**5.14 ms** (4.75) M 43.6 ms (41.4) **×0.12** IBP 1.61 ms	**5.14 ms** (4.75) αL 39.7 ms (36.8) **×0.13**	**4.80 ms** (3.90) cold 5.14 ms **×0.93**	**21.7 ms** (18.2) M 301 ms (300) **×1/14**	**21.7 ms** (18.2) αL 71.3 ms (68.6) **×0.30**	**31** digits
Tadpole T1 power 5 · rank 4	1	**4.26 ms** (4.26) M 22.3 ms (22.2) **×0.19** IBP 1.81 ms	**4.26 ms** (4.26) αL 22.9 ms (22.8) **×0.19**	**4.33 ms** (3.40) cold 4.26 ms **×1.0**	**12.2 ms** (11.0) M 154 ms (153) **×1/13**	**12.2 ms** (11.0) αL 32.1 ms (31.4) **×0.38**	**31** digits
Two loops							
QCD ladder, q = 0 1 gluon rung	2	**22.0 ms** (21.5) M 35.9 ms (34.1) **×0.61** IBP 5.87 ms	**22.0 ms** (21.5) αL 29.9 ms (29.9) **×0.74**	**18.9 ms** (18.4) cold 22.0 ms **×0.86**	**118 ms** (110) M 211 ms (204) **×0.56**	**118 ms** (110) αL 65.8 ms (65.0) **×1.8**	**31** digits
QCD ladder, O(q²) 1 gluon rung	2	**146 ms** (132) M 281 ms (278) **×0.52** IBP 9.65 ms	**146 ms** (132) αL 225 ms (197) **×0.65**	**139 ms** (134) cold 146 ms **×0.96**	**243 ms** (207) M 1.12 s (1.11) **×0.22**	**243 ms** (207) αL 299 ms (292) **×0.81**	**31** digits
Sunset T111 powers 3,2,2 · rank 4	2	**5.15 ms** (4.88) M 33.6 ms (30.8) **×0.15** IBP 1.30 ms	**5.15 ms** (4.88) αL 26.1 ms (24.5) **×0.20**	**5.18 ms** (4.89) cold 5.15 ms **×1.0**	**20.0 ms** (19.7) M 172 ms (172) **×0.12**	**20.0 ms** (19.7) αL 35.4 ms (33.2) **×0.57**	**31** digits
Three loops							
QCD ladder, q = 0 2 gluon rungs	14	**87.7 ms** (84.2) M 182 ms (181) **×0.48** IBP 65.5 ms	**87.7 ms** (84.2) αL 162 ms (162) **×0.54**	**49.4 ms** (49.3) cold 87.7 ms **×0.56**	**1.81 s** (1.80) M 1.10 s (1.08) **×1.6**	**1.81 s** (1.80) αL 277 ms (276) **×6.5**	**31** digits
QCD ladder, O(q²) 2 gluon rungs	14	**679 ms** (644) M 4.26 s (4.24) **×0.16** IBP 257 ms	**679 ms** (644) αL 4.71 s (4.69) **×0.14**	**486 ms** (483) cold 679 ms **×0.72**	**2.77 s** (2.65) M 11.9 s (11.7) **×0.23**	**2.77 s** (2.65) αL 5.01 s (4.85) **×0.55**	**31** digits
Tetrahedron D6 master	1	**12.0 ms** (11.8) M 68.4 ms (66.9) **×0.18** IBP 1.03 ms	**12.0 ms** (11.8) αL 26.4 ms (26.3) **×0.45**	**7.40 ms** (7.37) cold 12.0 ms **×0.62**	**25.9 ms** (25.2) M 252 ms (251) **×0.10**	**25.9 ms** (25.2) αL 59.5 ms (55.3) **×0.44**	**31** digits
Tetrahedron D6 3 dots · rank 4	38	**128 ms** (122) M 82.4 ms (82.3) **×1.6** IBP 118 ms	**128 ms** (122) αL 137 ms (136) **×0.93**	**72.7 ms** (71.3) cold 128 ms **×0.57**	**5.67 s** (5.52) M failed	**5.67 s** (5.52) αL 194 ms (194) **×29**	**31** digits
Four loops							
QCD ladder, q = 0 3 gluon rungs	10	**1.10 s** (1.02) F 5.84 s (5.82) **×0.19** IBP 336 ms	n/a	**924 ms** (864) cold 1.10 s **×0.84**	**7.44 s** (7.00) F 6.07 s (5.94) **×1.2**	n/a	**26** digits
QCD ladder, O(q²) 3 gluon rungs	10	**53.2 s** (52.0) F 467 s (457) **×0.11** IBP 1.50 s	n/a	**54.5 s** (54.3) cold 53.2 s **×1.0**	**61.9 s** (61.1) F 465 s (465) **×0.13**	n/a	**24** digits

Test case	Termi-nals	Reduction vs M / F	Reduction vs αL	Warm reduction	Full call vs M / F	Full call vs αL	Agree-ment
H family one line squared	15	**927 ms** (896) F 609 ms (602) **×1.5** IBP 853 ms	n/a	**433 ms** (413) cold 927 ms **×0.47**	**1.29 s** (1.26) F 632 ms (628) **×2.0**	n/a	**22** digits
FG family one line to the 4th	15	**225 ms** (224) F 366 ms (365) **×0.61** IBP 169 ms	n/a	**121 ms** (117) cold 225 ms **×0.54**	**2.70 s** (2.68) F 403 ms (399) **×6.7**	n/a	**23** digits
X family one line to the 4th	19	**4.86 s** (4.82) F 17.8 s (17.6) **×0.27** IBP 4.66 s	n/a	**3.04 s** (2.99) cold 4.86 s **×0.63**	**5.73 s** (5.57) F 17.8 s (17.7) **×0.32**	n/a	**21** digits
Five loops							
QCD ladder, q = 0 4 gluon rungs	79	**60.2 s**† (55.9) no FORM backend on 8 cores	n/a	**34.0 s**† (32,5) cold 60.2 s† **×0.57** on 8 cores	**3.49 h**† 1 run no FORM backend on 8 cores	n/a	—
QCD ladder, O(q²) 4 gluon rungs	79	**15.2 min**† 1 run no FORM backend on 8 cores	n/a	**574 s**† 1 run cold 15.2 min† **×0.63** on 8 cores	**6.16 h**† 1 run no FORM backend on 8 cores	n/a	—
Top-sector master (P105) 12 lines · scalar	1	**151 ms** (143) no FORM backend IBP 642 µs	n/a	**156 ms** (144) cold 151 ms **×1.0**	**154 ms** (144) no FORM backend	n/a	—
Dotted master (P63) 10 lines · rank 1	1	**144 ms** (137) no FORM backend IBP 599 µs	n/a	**130 ms** (127) cold 144 ms **×0.90**	**347 ms** (326) no FORM backend	n/a	—
Top sector (P105) master + one dot	98	**503 s**† 1 run no FORM backend on 8 cores	n/a	**300 s** 1 run cold 503 s† **×0.60** on 8 cores	failed no FORM backend	n/a	—
Dotted-master sector (P63) master + one dot	79	**54.8 s** (54.2) no FORM backend on 8 cores	n/a	**25.8 s** (25.5) cold 54.8 s **×0.47** on 8 cores	failed no FORM backend	n/a	—



† **Median not timed in an idle window** (the other cells' medians were, some after re-measurement): 5L QCD ladder, q = 0 · reduction: other tasks used 29.8% of each of the 8 timed CPUs; 5L QCD ladder, q = 0 · warm reduction: other tasks used 14.2% of each of the 8 timed CPUs; 5L QCD ladder, q = 0 · full call: one 3.5 h run (2026-11-06 22:44Z to 11-07 02:14Z) alongside other benchmark lanes of ours on other CPUs; other tasks used 21.0% of each of the 8 timed CPUs; 5L QCD ladder, O(q²) · reduction: other tasks used 4.0% of each of the 8 timed CPUs; 5L QCD ladder, O(q²) · warm reduction: other tasks used 4.0% of each of the 8 timed CPUs; 5L QCD ladder, O(q²) · full call: one 6.2 h run (2026-11-07 02:14Z to 08:25Z) alongside other benchmark and integration lanes of ours on other CPUs; 5L Top sector · reduction: other tasks used 7.7% of each of the 8 timed CPUs.

Once per process (excluded from every cell; first-call setup = first minus second call on the warm-up integral)

	1L	2L	3L	4L	5L
RustRed data load	1.85 s	1.82 s	869 ms	8.05 s	39.0 s
RustRed full call	1.93 ms	2.63 ms	22.6 ms	39.7 ms	862 ms
MATAD full call	928 ms	941 ms	870 ms	—	—
alphaLoop full call	892 ms	897 ms	809 ms	—	—
FMFT full call	—	—	—	813 ms	—

Failed calls.
3L Tetrahedron D6, 3 dots · rank 4 · MATAD full call (3/3 runs): vakint's precision guard rejects the backend's result: the imaginary part of the coefficient of 1/(mUVsq*ε^2) is −1.9e−71 ± 1.2e−70 (consistent with zero, no reliable digit)
5L Top sector (P105), master + one dot · full call (1/1 runs): vakint fails closed (MissingMasterOrders): 72 masters lack Laurent orders this result needs (the reduction coefficients have poles at d = 4; deepest orders needed: 5L to ε^1, 4L to ε^4, 3L to ε^3, 1L to ε^6), so the result fails from ε^−2 on
5L Dotted-master sector (P63), master + one dot · full call (1/1 runs): vakint's precision guard rejects the backend's result: the imaginary part of the coefficient of 1/(mUV^2*ε^3) is −2.2e−71 ± 3.0e−69 (consistent with zero, no reliable digit)

Provenance

Data (timings)	vakint-external-data `v0.2.1` · commit `fb978fb9aa55` · root manifest `87715e98cd999ca0`
Data (tables)	vakint-external-data v0.2.2 (0faa1c795664): precision tables regenerated from this later release, whose reduction payloads equal the timed tree's (receipt rule 2)
Vakint	0.2.0 · release build of `71316be4ff16` (clean) before the rebase onto the data pin; same source as `4852765c8259` on this branch outside docs and vendored data (receipt rule 1) · harness `951c0005eac19098` · FORM 5.0.0
Host	AMD EPYC 9754 128-Core Processor, 384 logical CPUs, shared; timed on 42 pinned CPUs in 9 L3 domains (listed in suite.json)
Contention	318 samples, 12 still contended after re-measurement († thresholds per timed CPU: SMT sibling 25% busy, other tasks 3%, call run-queue wait 3% when other tasks ran) · medians (max): SMT sibling busy 2.4% (23.3%), timed call in the run queue 0.2% (11%), other tasks per timed CPU 1.2% (47.3%) · 1-min load ≤ 294.6
Regenerate	`crates/vakint/docs/benchmark/make-benchmark.sh DATA_ROOT BINARY OUT --install` after a data release `DATA_ROOT - OUT --refresh-tables --install` (timings kept) · assembled 2026-11-07T11:37:06Z

Reduction graphs

Nodes are the vacuum topologies of the shipped reduction, on the row of their number of lines; an edge removes one line (thicker: several lines), as in Fig. 21 of B. Ruijl's thesis (Leiden, 2017). Products (one-vertex-reducible graphs) are merged by their factors; numbers count masters (one to three loops: RustRed terminals).

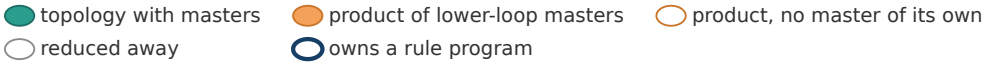




Figure 1: **1-loop reduction graph.** T1 carries the one-loop master; dots and numerators reduce inside it. 1 topology · 0 edges · 1 terminal.



Figure 2: **2-loop reduction graph.** Removing any sunset line gives T1². 2 topologies · 1 edge · 2 terminals.

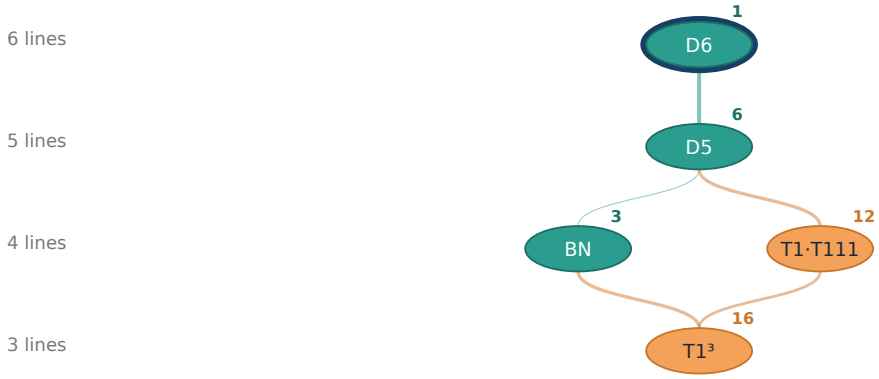


Figure 3: **3-loop reduction graph.** MATAD names; symmetric sectors are separate terminals of the closed program. 5 topologies · 5 edges · 38 terminals.

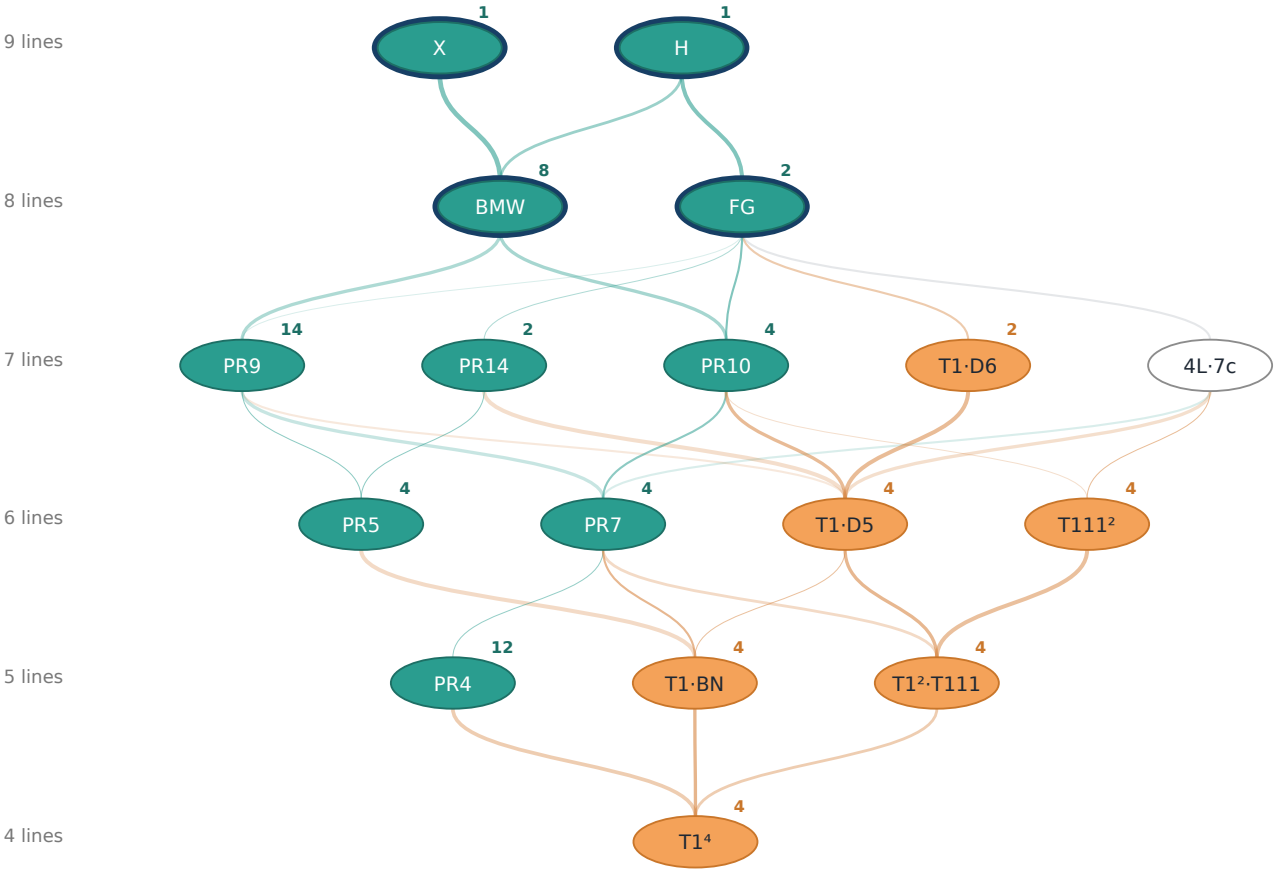


Figure 4: **4-loop reduction graph.** Rings: the FMFT family programs H, X, BMW, FG; irreducible topologies named by their FMFT master PRk. 17 topologies · 32 edges · 74 masters.

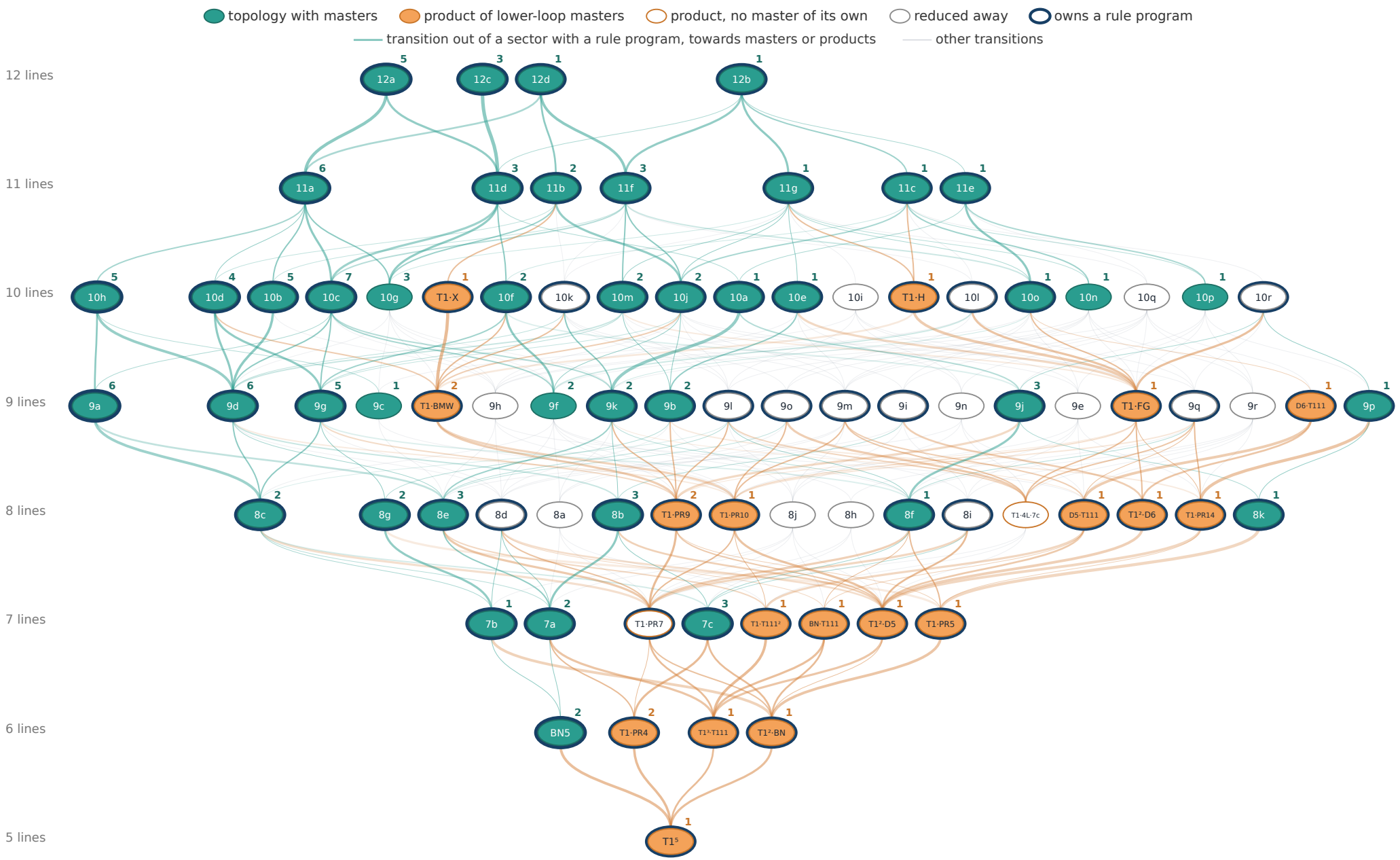


Figure 5: **5-loop reduction graph.** Rings: the 67 owner sectors of the saved application (width grows with their rule tables); names: lines and a letter. Benchmark masters: P105 in 12b, P63 in 10c. 82 topologies · 311 edges · 131 masters.

Precision of the masters and of the results

FINAL · data tree vakint-external-data v0.2.2 (0faa1c795664) (git 0faa1c795664, tree b738993f1fe0) · v0.2.2 root manifest 5ce0e4a0abe8687c · 4L constants 1338cfe501ffb516 · 5L values build 2026-11-07T02:24:15Z (producer-update 781d858170cd9127) · generated 2026-11-07T11:37:19Z · tables 0e2fc8ca25f526ec

Class of every Laurent coefficient, worst case X rounded down. **Precision table 1**: every master the data tree ships, at every order it holds. **Precision table 2**: the result of any single-scale vacuum input with L loops (RustRed backend), worst case over the inputs; an order that needs master coefficients the data do not hold (spurious poles of the reduction, lower-loop factors of 5L products) is not available.

S exact, periods at any precision · **S*(X)** exact, periods to at most X digits · **N(X)** numerical only, at most X digits · — not available

Precision table 1. Master coefficients — every master of the data tree (rows, by loop count; 5L: the 640-pack name in brackets) at every Laurent order held (columns); — = not held. [p] X rests in part on a constant computed with two profiles of one code beyond its independent confirmation; [n] Numerical mode returns more digits.

master	ε^{-5}	ε^{-4}	ε^{-3}	ε^{-2}	ε^{-1}	ε^{0}	ε^{1}	ε^{2}	ε^{3}	ε^{4}	ε^{5}
1L · 1 master · 7 S, 0 S*, 0 N											
1L/P1	—	—	—	—	S	S	S	S	S	S	S
2L · 2 masters · 13 S, 0 S*, 0 N											
2L/P1	—	—	—	S	S	S	S	S	S	—	—
2L/P2	—	—	—	S	S	S	S	S	S	S	—
3L · 38 masters · 252 S, 0 S*, 0 N											
3L/P1	—	—	S	S	S	S	S	S	—	—	—
3L/P2	—	—	S	S	S	S	S	S	—	—	—
3L/P3	—	—	S	S	S	S	S	S	—	—	—
3L/P4	—	—	S	S	S	S	S	S	S	—	—
3L/P5	—	—	S	S	S	S	S	S	—	—	—
3L/P6	—	—	S	S	S	S	S	S	—	—	—
3L/P7	—	—	S	S	S	S	S	S	—	—	—
3L/P8	—	—	S	S	S	S	S	S	S	—	—
3L/P9	—	—	S	S	S	S	S	S	—	—	—
3L/P10	—	—	S	S	S	S	S	S	S	—	—
3L/P11	—	—	S	S	S	S	S	S	—	—	—
3L/P12	—	—	S	S	S	S	S	S	S	—	—
3L/P13	—	—	S	S	S	S	S	S	S	S	—
3L/P14	—	—	S	S	S	S	S	S	S	—	—
3L/P15	—	—	S	S	S	S	S	S	—	—	—
3L/P16	—	—	S	S	S	S	S	S	—	—	—
3L/P17	—	—	S	S	S	S	S	S	—	—	—
3L/P18	—	—	S	S	S	S	S	S	S	—	—
3L/P19	—	—	S	S	S	S	S	S	—	—	—
3L/P20	—	—	S	S	S	S	S	S	—	—	—
3L/P21	—	—	S	S	S	S	S	S	S	S	—
3L/P22	—	—	S	S	S	S	S	S	S	—	—
3L/P23	—	—	S	S	S	S	S	S	S	—	—
3L/P24	—	—	S	S	S	S	S	S	S	—	—
3L/P25	—	—	S	S	S	S	S	S	—	—	—
3L/P26	—	—	S	S	S	S	S	S	S	—	—
3L/P27	—	—	S	S	S	S	S	S	—	—	—
3L/P28	—	—	S	S	S	S	S	S	S	S	—
3L/P29	—	—	S	S	S	S	S	S	S	—	—
3L/P30	—	—	S	S	S	S	S	S	S	—	—
3L/P31	—	—	S	S	S	S	S	S	—	—	—
3L/P32	—	—	S	S	S	S	S	S	S	—	—
3L/P33	—	—	S	S	S	S	S	S	S	—	—
3L/P34	—	—	S	S	S	S	S	S	S	—	—
3L/P35	—	—	S	S	S	S	S	S	S	—	—
3L/P36	—	—	S	S	S	S	S	S	S	—	—
3L/P37	—	—	S	S	S	S	S	S	S	—	—
3L/P38	—	—	S	S	S	S	S	S	—	—	—
4L · 74 masters · 376 S, 92 S*, 0 N											
4L/P1	—	S	S	S	S	S	S	S	—	—	—
4L/P2	—	S	S	S	S	S	S	S	S	—	—
4L/P3	—	S	S	S	S	S	S	S	—	—	—
4L/P4	—	S	S	S	S	S	S	S	—	—	—
4L/P5	—	S	S	S	S	S	S	—	—	—	—
4L/P6	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P7	—	S	S	S	S	S*(20000)	S*(5542)	—	—	—	—
4L/P8	—	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—
4L/P9	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—

master	ε^{-5}	ε^{-4}	ε^{-3}	ε^{-2}	ε^{-1}	ε^{0}	ε^{1}	ε^{2}	ε^{3}	ε^{4}	ε^{5}
4L/P10	—	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—
4L/P11	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P12	—	S	S	S	S	S	S	S	—	—	—
4L/P13	—	S	S	S	S	S*(5542)	S*(5406)	—	—	—	—
4L/P14	—	S	S	S	S	S*(5358)	S*(5357)[p]	—	—	—	—
4L/P15	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P16	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P17	—	S	S	S	S	S*(5363)[n]	S*(5358)	—	—	—	—
4L/P18	—	S	S	S	S	S*(2395)[p]	S*(2395)[p]	—	—	—	—
4L/P19	—	S	S	S	S	S*(5363)[n]	S*(5361)	—	—	—	—
4L/P20	—	S	S	S	S	S	S	S	—	—	—
4L/P21	—	S	S	S	S	S	S	S	S	—	—
4L/P22	—	S	S	S	S	S	S	S	—	—	—
4L/P23	—	S	S	S	S	S	S	S	—	—	—
4L/P24	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P25	—	S	S	S	S	S*(20000)	S*(5542)	—	—	—	—
4L/P26	—	S	S	S	S	S	S	S	—	—	—
4L/P27	—	S	S	S	S	S*(5542)	S*(5406)	—	—	—	—
4L/P28	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P29	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P30	—	S	S	S	S	S	S	—	—	—	—
4L/P31	—	S	S	S	S	S	S	—	—	—	—
4L/P32	—	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—
4L/P33	—	S	S	S	S	S	S	—	—	—	—
4L/P34	—	S	S	S	S	S*(2805)[p]	S*(2601)[p]	—	—	—	—
4L/P35	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P36	—	S	S	S	S	S	S	S	—	—	—
4L/P37	—	S	S	S	S	S	S	S	S	—	—
4L/P38	—	S	S	S	S	S	S	S	—	—	—
4L/P39	—	S	S	S	S	S	S	S	—	—	—
4L/P40	—	S	S	S	S	S	S	—	—	—	—
4L/P41	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P42	—	S	S	S	S	S*(20000)	S*(5542)	—	—	—	—
4L/P43	—	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—
4L/P44	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P45	—	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—
4L/P46	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P47	—	S	S	S	S	S	S	S	—	—	—
4L/P48	—	S	S	S	S	S*(5542)	S*(5406)	—	—	—	—
4L/P49	—	S	S	S	S	S*(5358)	S*(5357)[p]	—	—	—	—
4L/P50	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P51	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P52	—	S	S	S	S	S*(5363)[n]	S*(5358)	—	—	—	—
4L/P53	—	S	S	S	S	S	S	S	—	—	—
4L/P54	—	S	S	S	S	S	S	S	S	—	—
4L/P55	—	S	S	S	S	S	S	S	—	—	—
4L/P56	—	S	S	S	S	S	S	S	—	—	—
4L/P57	—	S	S	S	S	S	S	—	—	—	—
4L/P58	—	S	S	S	S	S	S	—	—	—	—
4L/P59	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P60	—	S	S	S	S	S*(20000)	S*(5542)	—	—	—	—
4L/P61	—	S	S	S	S	S	S	S	—	—	—
4L/P62	—	S	S	S	S	S*(5542)	S*(5406)	—	—	—	—
4L/P63	—	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—

master	ε^{-5}	ε^{-4}	ε^{-3}	ε^{-2}	ε^{-1}	ε^{0}	ε^{1}	ε^{2}	ε^{3}	ε^{4}	ε^{5}
4L/P64	—	S	S	S	S	S	S	—	—	—	—
4L/P65	—	S	S	S	S	S*(2805)[p]	S*(2601)[p]	—	—	—	—
4L/P66	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P67	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P68	—	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—
4L/P69	—	S	S	S	S	S*(5358)	S*(5357)[p]	—	—	—	—
4L/P70	—	S	S	S	S	S*(2600)[p]	S*(2348)[p]	—	—	—	—
4L/P71	—	S	S	S	S	S*(5542)	S*(5527)	—	—	—	—
4L/P72	—	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—
4L/P73	—	S	S	S	S	S*(5363)[n]	S*(5361)	—	—	—	—
4L/P74	—	S	S	S	S	S*(5363)[n]	S*(5358)	—	—	—	—
5L · 131 masters · 562 S, 29 S*, 195 N											
5L/P1 (P21)	S	S	S	N(67)	N(58)	N(59)	—	—	—	—	—
5L/P2 (P29)	S	S	S	N(83)	N(83)	N(75)	—	—	—	—	—
5L/P3 (P33)	S	S	S	N(69)	N(64)	N(61)	—	—	—	—	—
5L/P4 (P37)	S	S	S	N(65)	N(58)	N(56)	—	—	—	—	—
5L/P5 (P49)	S	S	S	N(65)	N(58)	N(57)	—	—	—	—	—
5L/P6 (P60)	S	S	N(65)	N(65)	N(57)	N(57)	—	—	—	—	—
5L/P7 (P65)	S	S	S	S	S	S	—	—	—	—	—
5L/P8 (P67)	S	S	S	S	S	N(142)	—	—	—	—	—
5L/P9 (P69)	S	S	S	S	S	N(142)	—	—	—	—	—
5L/P10 (P73)	S	S	S	N(59)	N(57)	N(56)	—	—	—	—	—
5L/P11 (P76)	S	S	S	S	S	S	—	—	—	—	—
5L/P12	S	S	S	S	S	S	—	—	—	—	—
5L/P13	S	S	S	S	S	S	—	—	—	—	—
5L/P14 (P95)	S	S	S	S	S	N(57)	—	—	—	—	—
5L/P15 (P100)	S	S	S	S	S	N(59)	—	—	—	—	—
5L/P16 (P101)	S	S	S	S	S	S	—	—	—	—	—
5L/P17 (P111)	S	S	S	N(59)	N(56)	N(56)	—	—	—	—	—
5L/P18 (P115)	S	S	S	N(65)	N(62)	N(58)	—	—	—	—	—
5L/P19 (P124)	S	S	S	S	N(57)	N(58)	—	—	—	—	—
5L/P20 (P126)	S	S	S	S	N(65)	N(61)	—	—	—	—	—
5L/P21 (P129)	S	S	S	S	S*(5542)	S*(5406)	—	—	—	—	—
5L/P22 (P131)	S	S	S	N(65)	N(57)	N(56)	—	—	—	—	—
5L/P23 (P134)	S	S	S	S	N(84)	N(75)	—	—	—	—	—
5L/P24 (P140)	S	S	S	S	N(57)	N(57)	—	—	—	—	—
5L/P25 (P142)	S	S	S	S	S	N(60)	—	—	—	—	—
5L/P26 (P144)	S	S	S	S	S*(5363)	N(57)	—	—	—	—	—
5L/P27 (P158)	S	S	S	S	S	S	—	—	—	—	—
5L/P28 (P159)	S	S	S	S	S	S	—	—	—	—	—
5L/P29 (P169)	S	S	S	S	N(56)	N(56)	—	—	—	—	—
5L/P30 (P170)	S	S	S	S	S	N(47)	—	—	—	—	—
5L/P31 (P171)	S	S	S	S	S	N(48)	—	—	—	—	—
5L/P32 (P172)	S	S	S	S	S	N(47)	—	—	—	—	—
5L/P33 (P178)	S	S	S	S	N(83)	N(75)	—	—	—	—	—
5L/P34 (P179)	S	S	S	S	N(83)	N(75)	—	—	—	—	—
5L/P35 (P182)	S	S	S	S	S*(2805)[p]	S*(2601)[p]	—	—	—	—	—
5L/P36 (P202)	S	S	S	S	S	N(69)	—	—	—	—	—
5L/P37 (P204)	S	S	S	S	S	N(56)	—	—	—	—	—
5L/P38 (P238)	S	S	S	S	N(56)	N(55)	—	—	—	—	—
5L/P39 (P245)	S	S	S	S	S*(5363)	N(57)	—	—	—	—	—
5L/P40 (P263)	S	S	S	S	N(55)	N(56)	—	—	—	—	—
5L/P41 (P264)	S	S	S	S	S	N(43)	—	—	—	—	—
5L/P42 (P299)	S	S	S	S	S	N(58)	—	—	—	—	—
5L/P43 (P312)	S	S	S	S	N(57)	N(57)	—	—	—	—	—
5L/P44 (P314)	S	S	S	S	S	N(74)	—	—	—	—	—
5L/P45 (P315)	S	S	S	S	S	N(66)	—	—	—	—	—
5L/P46 (P321)	S	S	S	S	N(57)	N(58)	—	—	—	—	—
5L/P47 (P352)	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—	—
5L/P48 (P353)	S	S	S	S	S	S	—	—	—	—	—
5L/P49 (P356)	S	S	S	S	S	S	—	—	—	—	—
5L/P50 (P357)	S	S	S	N(47)	N(47)	N(47)	—	—	—	—	—
5L/P51 (P359)	S	S	S	S	S*(20000)	S*(20000)	—	—	—	—	—
5L/P52 (P361)	S	S	S	N(65)	N(57)	N(57)	—	—	—	—	—
5L/P53 (P367)	S	S	S	S	N(83)	N(75)	—	—	—	—	—
5L/P54 (P368)	S	S	S	S	N(82)	N(76)	—	—	—	—	—
5L/P55 (P369)	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—	—
5L/P56 (P370)	S	S	S	S	N(82)	N(74)	—	—	—	—	—
5L/P57 (P375)	S	S	S	S	N(82)	N(76)	—	—	—	—	—
5L/P58 (P379)	S	S	S	S	S	N(74)	—	—	—	—	—
5L/P59 (P382)	S	S	S	S	S*(5358)	S*(5357)[p]	—	—	—	—	—
5L/P60 (P383)	S	S	S	N(65)	N(57)	N(57)	—	—	—	—	—
5L/P61 (P384)	S	S	S	S	S	N(57)	—	—	—	—	—
5L/P62 (P385)	S	S	S	S	S	N(56)	—	—	—	—	—
5L/P63 (P387)	S	S	S	N(60)	N(56)	N(51)	—	—	—	—	—
5L/P64 (P389)	S	S	S	S	S	N(56)	—	—	—	—	—
5L/P65 (P391)	S	S	S	S	S	N(56)	—	—	—	—	—
5L/P66 (P398)	S	S	S	N(66)	N(57)	N(59)	—	—	—	—	—

master	ε^{-5}	ε^{-4}	ε^{-3}	ε^{-2}	ε^{-1}	ε^{0}	ε^{1}	ε^{2}	ε^{3}	ε^{4}	ε^{5}
5L/P67 (P399)	S	S	S	S	S	N(57)	—	—	—	—	—
5L/P68 (P400)	S	S	S	S	S	N(3)	—	—	—	—	—
5L/P69 (P413)	S	S	S	S	N(83)	N(74)	—	—	—	—	—
5L/P70 (P418)	S	S	S	S	N(82)	N(74)	—	—	—	—	—
5L/P71	S	S	S	S	N(57)	N(58)	—	—	—	—	—
5L/P72	S	S	S	N(64)	N(57)	N(57)	—	—	—	—	—
5L/P73 (P422)	S	S	S	N(82)	N(82)	N(74)	—	—	—	—	—
5L/P74 (P424)	S	S	S	S	N(84)	N(75)	—	—	—	—	—
5L/P75 (P426)	S	S	S	S	N(58)	N(57)	—	—	—	—	—
5L/P76 (P427)	S	S	S	S	N(83)	N(75)	—	—	—	—	—
5L/P77 (P428)	S	S	S	S	S	N(57)	—	—	—	—	—
5L/P78 (P429)	S	S	S	S	S	N(74)	—	—	—	—	—
5L/P79 (P432)	S	S	S	S	S	N(62)	—	—	—	—	—
5L/P80 (P433)	S	S	S	S	S*(2395)[p]	N(47)	—	—	—	—	—
5L/P81	S	S	S	N(65)	N(57)	N(57)	—	—	—	—	—
5L/P82 (P435)	S	S	S	N(48)	N(47)	N(47)	—	—	—	—	—
5L/P83 (P436)	S	S	S	S	N(82)	N(75)	—	—	—	—	—
5L/P84 (P440)	S	S	S	N(47)	N(47)	N(47)	—	—	—	—	—
5L/P85 (P441)	S	S	S	S	S	N(3)	—	—	—	—	—
5L/P86 (P442)	S	S	S	S	S*(2395)[p]	S*(2395)[p]	—	—	—	—	—
5L/P87 (P443)	S	S	S	S	S	N(57)	—	—	—	—	—
5L/P88 (P444)	S	S	S	S	S	N(44)	—	—	—	—	—
5L/P89 (P445)	S	S	S	S	S	N(3)	—	—	—	—	—
5L/P90 (P446)	S	S	S	S	S	N(35)	—	—	—	—	—
5L/P91 (P447)	S	S	S	S	S	S	—	—	—	—	—
5L/P92 (P448)	S	S	S	S	N(57)	N(57)	—	—	—	—	—
5L/P93 (P449)	S	S	S	S	S	S	—	—	—	—	—
5L/P94 (P451)	S	S	S	S	N(57)	N(57)	—	—	—	—	—
5L/P95 (P453)	S	S	S	S	N(83)	N(75)	—	—	—	—	—
5L/P96 (P454)	S	S	S	S	S*(2600)[p]	S*(2348)[p]	—	—	—	—	—
5L/P97 (P455)	S	S	S	N(47)	N(47)	N(47)	—	—	—	—	—
5L/P98 (P456)	S	S	S	S	S*(2600)[p]	N(47)	—	—	—	—	—
5L/P99 (P457)	S	S	S	S	S	N(53)	—	—	—	—	—
5L/P100 (P460)	S	S	S	S	N(82)	N(75)	—	—	—	—	—
5L/P101 (P462)	S	S	N(48)	N(48)	N(48)	N(48)	—	—	—	—	—
5L/P102 (P463)	S	S	S	N(47)	N(47)	N(47)	—	—	—	—	—
5L/P103 (P464)	S	S	S	S	S	N(48)	—	—	—	—	—
5L/P104 (P465)	S	S	S	S	S*(2600)[p]	N(47)	—	—	—	—	—
5L/P105 (P466)	S	S	S	S	S	N(55)	—	—	—	—	—
5L/P106 (P470)	S	S	S	S	N(82)	N(74)	—	—	—	—	—
5L/P107 (P472)	S	S	S	S	N(47)	N(46)	—	—	—	—	—
5L/P108 (P477)	S	S	S	S	N(57)	N(57)	—	—	—	—	—
5L/P109 (P478)	S	S	S	S	S*(5363)	N(75)	—	—	—	—	—
5L/P110 (P480)	S	S	S	S	N(82)	N(75)	—	—	—	—	—
5L/P111 (P483)	S	S	S	S	S	N(56)	—	—	—	—	—
5L/P112 (P484)	S	S	S	S	S	N(55)	—	—	—	—	—
5L/P113	S	S	S	S	N(56)	N(56)	—	—	—	—	—
5L/P114 (P501)	S	S	S	S	S*(5363)	N(56)	—	—	—	—	—
5L/P115 (P505)	S	S	S	S	S	N(43)	—	—	—	—	—
5L/P116 (P506)	S	S	S	S	S*(5363)	N(57)	—	—	—	—	—
5L/P117 (P509)	S	S	S	S	N(35)	N(35)	—	—	—	—	—
5L/P118 (P511)	S	S	S	S	N(56)	N(57)	—	—	—	—	—
5L/P119 (P519)	S	S	S	S	S*(5527)	S*(5525)	—	—	—	—	—
5L/P120 (P530)	S	S	S	S	S*(5363)	S*(5361)	—	—	—	—	—
5L/P121 (P533)	S	S	S	S	N(82)	N(74)	—	—	—	—	—
5L/P122 (P536)	S	S	S	S	N(81)	N(74)	—	—	—	—	—
5L/P123 (P539)	S	S	S	S	S	N(62)	—	—	—	—	—
5L/P124 (P550)	S	S	S	S	S*(5363)	N(75)	—	—	—	—	—
5L/P125 (P551)	S	S	S	S	S	N(3)	—	—	—	—	—
5L/P126 (P553)	S	S	S	S	S	N(61)	—	—	—	—	—
5L/P127 (P560)	S	S	S	S	N(82)	N(74)	—	—	—	—	—
5L/P128	S	S	S	S	N(56)	N(56)	—	—	—	—	—
5L/P129 (P631)	S	S	S	S	S	N(63)	—	—	—	—	—
5L/P130 (P632)	S	S	S	S	S	N(3)	—	—	—	—	—
5L/P131 (P634)	S	S	S	S	S	N(54)	—	—	—	—	—

[p] Two-profile constants (digits; independent confirmation): vakint::D5bar3 (9924 d; independent implementations agree to 5060 d): 4 orders; vakint::PR0D2Eps1 (2395 d; independent confirmation 48.5 d): 5 orders; vakint::PR0D2Eps2 (2396 d; independent confirmation 51.2 d): 2 orders; vakint::PR12D2Eps2 (2600 d; independent confirmation 48.7 d): 6 orders; vakint::PR12D2Eps3 (2348 d; independent confirmation 421 d): 2 orders; vakint::PR15D2Eps1 (2805 d; independent confirmation 794.27 d): 6 orders; vakint::PR15D2Eps2 (2601 d; independent confirmation 794.42 d): 9 orders; vakint::PR15D2Eps3 (2351 d; independent confirmation 793.39 d): 2 orders; vakint::PR5D2Eps4 (9923 d; independent implementations agree to 5060 d): 4 orders. [n] Numerical mode uses the best payload of a master's coefficient group (same exact basis expression), here a validated numerical value: 4L/P17 ε^0 S*(5363) → 20000 d; 4L/P19 ε^0 S*(5363) → 20000 d; 4L/P52 ε^0 S*(5363) → 20000 d; 4L/P73 ε^0 S*(5363) → 20000 d; 4L/P74 ε^0 S*(5363) → 20000 d.

Precision table 2. The L-loop result — class and digits reachable, worst case over the admitted inputs, per Laurent order of the result, with the spurious-pole orders the row notes state (measured at 1L-4L; at 5L proven on the held integrals and sampled outside them, but not in the top sector: a measured top-sector input is folded in, note k; letters: notes below).

L	ε^{-5}	ε^{-4}	ε^{-3}	ε^{-2}	ε^{-1}	ε^{0}	ε^{1}	ε^{2}	ε^{3}	ε^{4}	ε^{5}
1L[a]					S	S	S	S	S	S	S
2L[a]				S	S	S	S	S	S	—[b] some inputs: S	—
3L[a]			S	S	S	S	S	—[c] some inputs: S	—[d] some inputs: S	—[e] some inputs: S	—
4L[a]		S	S	S	S* 20000 digits[f]	S* 2395 digits[g]	—[h] some inputs: S*(2348)	—[i] some inputs: S	—[j] some inputs: S	—	—
5L[k,l]	S	N 65 digits[m] others: S	N 47 digits[n] others: N(48)	—[o] some inputs: N(47)	—[p] some inputs: N(35)	—[q] some inputs: N(3)	—	—	—	—	—

a 1L-4L: spurious-pole orders p measured on the shipped caches and samples (direct route up to 1L 0, 2L 0, 3L 1, 4L 2), not proven.
b 2L ε^{4}: inputs reaching 2L/P1 get MissingMasterOrders; others S.
c 3L ε^{2}: inputs reaching 3L/P1, 3L/P3, 3L/P6 +9 get MissingMasterOrders; others S.
d 3L ε^{3}: inputs reaching 3L/P1, 3L/P2, 3L/P3 +14 get MissingMasterOrders; others S.
e 3L ε^{4}: inputs reaching 3L/P1, 3L/P2, 3L/P3 +35 get MissingMasterOrders; only inputs with lower spurious poles, e.g. 3L/P13, 3L/P21, 3L/P28: S.
f 4L ε^{-1} S*(20000): K5Moment(1), K5Moment(3) via $1/\varepsilon^{1}$ poles.
g 4L ε^{0} S*(2395): PR0D2Eps1 (two profiles of one code).
h 4L ε^{1}: inputs reaching 4L/P6, 4L/P7, 4L/P9 +18 get MissingMasterOrders; so do 4L/P5, 4L/P8, 4L/P15 +5 combined with 4L/P35, 4L/P51; others S*(2348) (4L/P70 ε^{1}.
i 4L ε^{2}: inputs reaching 4L/P1, 4L/P3, 4L/P5 +59 get MissingMasterOrders; others S.
j 4L ε^{3}: inputs reaching 4L/P1, 4L/P2, 4L/P3 +71 get MissingMasterOrders; only inputs with lower spurious poles, e.g. 4L/P2, 4L/P21, 4L/P37 +1: S.
k 5L: p = 0 (pole-free basis BASIS-v2) proven on the 2336 held integrals and sampled outside them. Not so in the top sector: the measured input P105 + dot has p = 1 on 59 5L masters and up to p = 3 on its 4L product factors; other top-sector inputs are not validated.
l 5L results hold products of 1L-4L masters: a factor of L_f loops is needed through ε^{k+5-L_f+p} (p measured on the factor images).
m 5L ε^{-4} N(65): P105 + dot needs 5L/P6 (P60) ε^{-3}; others S.
n 5L ε^{-3} N(47): P105 + dot needs 5L/P50 (P357) ε^{-2}, 5L/P84 (P440) ε^{-2}; others N(48).
o 5L ε^{-2}: P105 + dot gets MissingMasterOrders; others N(47) (5L/P102 (P463) ε^{-2}, 5L/P50 (P357) ε^{-2}.
p 5L ε^{-1}: P105 + dot gets MissingMasterOrders; others N(35) (5L/P117 (P509) ε^{-1}.
q 5L ε^{0}: P105 + dot gets MissingMasterOrders; others N(3) (5L/P125 (P551) ε^{0}, 5L/P130 (P632) ε^{0}.

Method. A coefficient is limited by its fewest-digit finite source (4L constants, numerical master values); the result at ε^{k} needs every master m through ε^{k+p_m} (p_m: spurious-pole order of its reduction coefficient), on vakint's basis-alias route first, else the direct route; a needed order the data do not hold gives MissingMasterOrders (no value, never a wrong one). RustRed backend only. Full notes, method, finite-digit periods, the numerical identities behind some S and the PSLQ identifications: `crates/vakint/docs/benchmark/data/PRECISION-TABLES.md`.